Telecommunications Engineering Modular Analysis of Analog FM and Digital PCM Architectures

Part 1: Frequency Modulation & Demodulation

Q1: Why does the carrier frequency change in an FM system?

The frequency is dynamic because the **Voltage Controlled Oscillator (VCO)** acts as a transducer that converts input voltage into frequency. As the message signal's amplitude fluctuates, it shifts the oscillator's timing, causing the output frequency to deviate from its rest state.

Q2: What is the relationship between frequency deviation and the message?

The frequency deviation is **directly proportional** to the message **amplitude**. However, it is independent of the message frequency; the message frequency only determines how fast the carrier "swings" between its frequency limits.

Q3: How does a Zero-Crossing Detector (ZCD) recover the message?

The ZCD triggers a fixed-width pulse every time the FM signal crosses 0V. This creates a pulse train where the **duty cycle** represents the original signal's frequency. A Low-Pass Filter then averages this duty cycle to reconstruct the analog message.

Part 2: Sampling & Reconstruction

Q1: What is the significance of the Nyquist Rate?

The Nyquist Rate ($f_s > 2f_m$) is the minimum sampling frequency required to fully capture a signal. If you sample below this rate, information is lost and "aliasing" occurs, creating ghost frequencies that distort the output.

Q2: Why is a Low-Pass Filter (LPF) necessary for reconstruction?

Sampling creates high-frequency replicas (images) of the signal. The LPF acts as an anti-imaging tool, removing these high-frequency components and "smoothing" the discrete samples back into a continuous analog waveform.

Part 3: PCM Encoding & Decoding

Q1: What is "Quantization Error," and how does it manifest?

Quantization error is the difference between the actual analog voltage and the nearest digital level. In your 4-bit system, this appears as a "staircase" effect on the reconstructed wave, which creates a specific type of distortion known as **quantization noise**.

Q2: Why is Frame Synchronization vital for digital decoding?

Since a PCM bitstream is a continuous serial string of 1s and 0s, the decoder has no inherent way of knowing where one 4-bit word ends and the next begins. The **Frame Sync** pulse provides the timing reference needed to correctly group the bits into their original values.

Part 4: Bandwidth Limiting & Restoration

Q1: Why does a limited bandwidth "round off" a digital pulse?

A square pulse is made of a fundamental frequency and infinite high-frequency harmonics. A band-limited channel acts as a **Low-Pass Filter**, stripping away these harmonics and leaving only the low-frequency "rounded" shape.

Q2: What is Intersymbol Interference (ISI)?

ISI occurs when the bandwidth is so restricted that one pulse "smears" into the time slot of the next. This overlap makes it difficult for the receiver to distinguish between consecutive bits.

Q3: How does the Decision Circuit (Comparator) restore the signal?

The circuit compares the rounded, noisy signal against a **DC threshold**. If the voltage is above the threshold, it regenerates a clean 'High' state; if below, a 'Low' state. This effectively ignores the rounding and noise to recreate the original square bitstream.